INSULIN GLARGINE

High-alert drug
Never mix or IV

CHEMICAL NAME A21-glycine, B31-arginine, B32-arginine human insulin (recombinant long-acting analogue)
CLASS Long-acting basal insulin analogue
TRADE NAMES Lantus · Toujeo (U-300) · Abasaglar · Semglee

THERAPEUTIC USES / INDICATIONS

- Basal insulin in type 1 (with prandial insulin) and type 2 diabetes (alone or with oral agents).
- Continue basal insulin during DKA treatment and in hospital.

ADMINISTRATION

- **SC once daily at the same time each day** (any time, consistently).
- Do not mix with other insulins; do not dilute; never IV.
- Clear solution (do not confuse with rapid-acting insulin — both clear).
- Toujeo U-300: pen only; dose not 1:1 with Lantus (often 10–18% more).

DOSAGES

- **Type 2 start:** 0.2 unit/kg or up to 10 units once daily, then titrate to the fasting glucose target.
- **Type 1:** start at about one-third of total daily insulin; rest as prandial insulin.
- Switching from twice-daily NPH: start at 80% of total NPH dose.

PHARMACOKINETICS

- **Onset:** 1–2 h. No pronounced peak. **Duration:** ~24 h (Toujeo up to 36 h).
- Precipitates at injection site (neutral pH) → slow release.

PHARMACODYNAMICS

- Steady basal insulin effect → suppresses hepatic glucose output between meals and overnight.
- Less nocturnal hypoglycaemia than NPH.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain (acidic solution), lipohypertrophy.
- Weight gain.

ADVERSE EFFECTS (SERIOUS)

- Hypoglycaemia (less nocturnal than NPH).
- Hypokalaemia.
- Medication errors (confusing with rapid-acting insulin).

CONTRAINDICATIONS & CAUTIONS

- Hypoglycaemia; hypersensitivity.
- Not for DKA treatment (IV soluble insulin needed); but continue SC basal.

KEY INTERACTIONS

- As other insulins.

NURSING MANAGEMENT

- Give at the same time daily; record time.
- Do not omit even if nil by mouth (may reduce dose by ~20% per protocol).
- Monitor fasting glucose for titration.
- Clearly label pen; check name against prescription (clear like rapid insulins).

PATIENT EDUCATION

- Inject at the same time every day; never mix with other insulins.
- Adjust dose by fasting glucose as taught.
- Do not stop when ill (sick-day rules).
- Keep in-use pen at room temperature up to 4 weeks.

KEY POINTS

- Long-acting basal analogue — peakless, ~24 h.
- Once daily, same time.
- Never mix or give IV.
- Continue in hospital and during DKA.

SOURCES CHECKED **[595]** Lantus Solostar, SC (US label) **[596]** Insulin glargine U-300 Max, SC (US label)
Full references p. 432–457